



Starting Finance Club USI
Quantitative Finance Division

**Commodity Assets as Portfolio Diversifiers:
A Twenty-Year Empirical Analysis with DCC-GARCH,
Markov Regime-Switching, and Machine Learning**

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Abstract

This paper examines the diversification properties of five commodity futures (copper, silver, gold, wheat, and natural gas) relative to US equities over the 2005–2025 period, utilizing a continuous sample of 1,057 weekly observations. We deploy a rigorous econometric toolkit: time-varying linkages are estimated via Dynamic Conditional Correlation (DCC-GARCH) built on GARCH(1,1) marginals, while macro-financial states are endogenously extracted using Markov regime-switching models (a two-state model for equities and a three-state specification for gold). To formally test the financialization hypothesis, we introduce a non-linear machine-learning layer. While linear Granger causality tests fail to show that the VIX leads commodity–equity integration ($p > 0.10$), indicating contemporaneous pricing of systemic shocks, a tuned Random Forest with SHAP (*SHapley Additive exPlanations*) attribution models one-week-ahead dynamic correlation (out-of-sample $R^2 = 0.763$). While global statistical metrics are dominated by autoregressive persistence (AR(1) $R^2 = 0.783$), SHAP analysis reveals the machine learning model’s true marginal value: timing structural regime breakouts via Hidden Markov Models probabilities and institutional hedging pressure (CFTC COT data), rather than lagged equity volatility. Portfolio analysis is extended via CVaR-minimising allocations under Ledoit–Wolf shrinkage and transaction-cost-aware optimisations. Our findings mathematically challenge treating commodities as a monolithic asset class: gold remains the sole genuine diversifier, exhibiting a uniquely low equity beta (0.141) and commanding an optimal 50.0% weight in the unpenalised tangent portfolio, lifting the Sharpe ratio from 0.582 to 0.846. The DCC-GARCH framework formally confirms the diversification paradox for procyclical assets like copper, while gold maintains a flat bear–bull spread (+0.003). Finally, under realistic transaction costs of 5 bps anchored to a $1/N$ prior, the optimiser dynamically shifts to favour highly liquid equities (24.3%) while actively penalising volatile components with heavy rebalancing drag, offering actionable guidelines for robust portfolio construction.

Data sources: *Yahoo Finance, Federal Reserve FRED, CFTC Disaggregated COT Report. Code and dataset available on GitHub.*

Code Availability: The complete Python implementation (Jupyter Notebook and charts) developed for this work is available on GitHub. *Repository URL:* https://github.com/botta-andrea/SFclubUSI-Commodity_Diversification

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1. Introduction

Modern portfolio theory, originating with Markowitz (1952), establishes that the risk of a diversified portfolio depends not only on the volatility of individual assets but critically on their conditional correlations. Assets exhibiting low or negative correlations with equities can reduce portfolio variance without proportionally sacrificing expected returns, structurally shifting the efficient frontier upward and to the left. This mathematical insight has motivated decades of research into the diversification properties of alternative asset classes, including real estate, hedge funds, private equity, and commodity futures.

Unlike traditional financial assets, commodities derive their value from physical supply and demand dynamics, such as harvest cycles, mining production, weather patterns, and geopolitical disruptions, that are theoretically orthogonal to corporate earnings and broad equity valuations. Seminal work by Gorton and Rouwenhorst (2006) documented that commodity futures historically exhibited equity-like returns with uniquely low correlation to stocks and bonds. However, the financialization hypothesis (Tang and Xiong, 2012) has challenged this traditional view: the massive influx of institutional capital and passive index products into commodity markets may have structurally amplified the co-movement between commodities and equities, potentially eroding the diversification benefits that originally justified their inclusion in institutional portfolios.

This paper contributes to this ongoing debate along four key methodological dimensions. First, we abandon static rolling windows to document the time-varying nature of commodity–equity linkages over a twenty-year period, utilising robust Dynamic Conditional Correlation (DCC-GARCH) estimators. Second, we identify macro-financial regimes endogenously via a two-state Markov switching model for equities and a three-state model for gold, allowing us to accurately quantify the regime-dependence of correlation dynamics. Third, we evaluate the portfolio-level impact of these commodities using classical Mean-Variance optimisation, robust CVaR-based tail-risk mitigation, and transaction-cost-aware allocations (incorporating a 5 bps turnover penalty). Fourth, we rigorously test the predictive limits of linear econometric estimators via Granger causality, and subsequently deploy an advanced non-linear Machine Learning layer, integrating Hidden Markov Models, a tuned Random Forest with SHAP feature attribution, and a Stacked Ensemble, to demonstrate that future correlation states are indeed forecastable from structural market signals.

The empirical analysis covers five heterogeneous commodity futures: copper, silver, gold, wheat, and natural gas. The core continuous dataset spans January 2005 to December 2025, yielding 1,057 weekly observations derived from COMEX, CBOT, and NYMEX closing prices, contextualised by macroeconomic data from the Federal Reserve (FRED). To explicitly test the impact of speculative hedging pressure, institutional positioning data from the CFTC Disaggregated Commitments of Traders (COT) reports are integrated for the valid machine-learning sub-sample of January 2013 to November 2024 (597 weekly observations).

Ultimately, by integrating econometric analysis with portfolio optimisation, this paper offers robust conclusions for academic research and practical insights for retail asset management.

2. Theoretical Framework

2.1 Commodity Pricing and Portfolio Theory

Commodity futures are fundamentally distinct from equities because they represent claims on physical assets rather than future cash flows. The total return of a commodity futures position decomposes into the spot price return, the collateral return, and the roll yield (Fama and French, 1987). According to the theory of storage and the hedging pressure hypothesis, commercial producers systematically short futures to hedge price risk, offering a risk premium to speculators who provide liquidity on the long side (De Roon et al., 2000). This structural imbalance naturally pushes futures markets into normal backwardation, generating a positive roll yield that is largely independent of broad equity market movements (Erb and Harvey, 2006). Furthermore, commodities inherently act as a hedge against inflation: since raw materials directly feed into the Consumer Price Index (CPI), their futures tend to perform well during macroeconomic regimes characterized by rising prices, a period when real equity returns are typically compressed (Bodie, 1983).

2.2 Financialization and Asset Heterogeneity

The early 2000s witnessed a structural paradigm shift in commodity markets, commonly termed “financialization” (Tang and Xiong, 2012). Index funds and institutional investors began allocating massive capital to commodity futures as a distinct, homogenous asset class (Basak and Pavlova, 2016). This structural increase in financial participation amplified cross-market integration, increasing the co-movement between commodities and traditional risk assets and potentially eroding historical diversification benefits (Büyükhahin and Robe, 2014).

However, despite this financialization, commodities cannot be treated uniformly in a portfolio context (Daskalaki and Skiadopoulos, 2011). Precious metals, particularly gold, function primarily as monetary assets and safe havens, famously exhibiting zero or negative conditional correlation with equities during severe market downturns (Baur and McDermott, 2010; Baur and Lucey, 2010). Conversely, industrial metals (e.g., copper) and energy commodities are heavily tied to the global business cycle, exhibiting strong procyclical characteristics that make their correlation with the S&P 500 highly sensitive to macroeconomic growth.

2.3 Econometric Modeling of Dynamic Linkages

2.3.1 Conditional Volatility and Dynamic Correlations

To properly capture the stylized facts of financial time series, namely volatility clustering and time-varying co-movements, we must abandon static unconditional metrics. We model the conditional variance of each asset using the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) framework (Bollerslev, 1986). The variance evolution for asset i follows a GARCH(1,1) process:

$$\sigma_{i,t}^2 = \omega_i + \alpha_i \epsilon_{i,t-1}^2 + \beta_i \sigma_{i,t-1}^2 \quad (1)$$

where α_i captures the impact of recent idiosyncratic shocks and β_i dictates the persistence of the volatility (Bollerslev, 1986).

Building on these marginal distributions, the dynamic interaction between commodities and traditional risk assets is evaluated via the Dynamic Conditional Correlation (DCC) estimator (Engle, 2002). The conditional covariance matrix H_t is defined as $H_t = D_t R_t D_t$, where D_t is the diagonal matrix of conditional standard deviations and R_t is the dynamic correlation matrix. The correlation evolution is governed by:

$$Q_t = (1 - a - b)\bar{Q} + a(u_{t-1}u_{t-1}^\top) + bQ_{t-1} \quad (2)$$

where u_t represents the standardized residuals and $\bar{Q}$ is the unconditional correlation matrix (Engle, 2002).

2.3.2 Regime-Switching Dynamics and the Diversification Paradox

The “diversification paradox” states that asset correlations are heavily asymmetric, structurally increasing during periods of high volatility or systemic risk, precisely when diversification is needed most (Longin and Solnik, 2001; Ang and J. Chen, 2002). To formally model this non-linear dependence, we incorporate Hidden Markov Models (HMM) (Rabiner, 1989). We assume the macro-financial environment operates under a latent state variable $S_t \in \{1, 2\}$, identifying underlying “Bull/Low-Volatility” and “Bear/High-Volatility” regimes (Hamilton, 1989). The transition between states is governed by a constant Markov transition matrix $P = [p_{ij}]$, where $p_{ij} = \mathbb{P}(S_t = j \mid S_{t-1} = i)$. Empirically, these latent states are extracted from S&P 500 weekly returns to estimate regime-specific means and volatilities. To capture its unique dynamics, a separate three-state model (including a “safe-haven” regime) is independently fitted to gold.

2.4 Robust Portfolio Allocation and Tail Risk

Standard Mean-Variance optimization (Markowitz, 1952) is notoriously sensitive to estimation errors in the sample covariance matrix $\hat{\Sigma}$, often leading to highly concentrated and unstable portfolio weights. To address this structural instability, we apply Ledoit-Wolf covariance shrinkage, which optimally pulls the sample covariance matrix toward a highly structured target matrix (e.g., constant correlation) with intensity $\delta \in (0, 1)$ (Ledoit and Wolf, 2004):

$$\hat{\Sigma}_{LW} = \delta F + (1 - \delta)\hat{\Sigma} \quad (3)$$

While shrinkage improves out-of-sample robustness, classical variance remains a symmetric risk measure. To specifically protect the portfolio against severe market drawdowns, we extend our allocation framework to minimize the Conditional Value-at-Risk (CVaR) (Rockafellar and Uryasev, 2000). For a given confidence level α (e.g., 95%), CVaR optimizes the expected loss strictly in the worst $1 - \alpha$ tail of the return distribution, providing superior downside protection for non-normal commodity returns:

$$CVaR_\alpha = \mathbb{E}[L \mid L \geq VaR_\alpha] \quad (4)$$

2.5 Predictive Modeling via Machine Learning

Given that financialization intimately links correlation regimes to macro-financial sentiment (Tang and Xiong, 2012), we hypothesize that future correlation structures can be anticipated. We utilize Granger causality tests (Granger, 1969) to establish lead-lag temporal relationships between broad risk proxies (e.g., the VIX) and commodity-equity correlation shifts. To capture these complex, non-linear predictive relationships, we deploy a stacked ensemble of machine learning algorithms, integrating Random Forests (Breiman, 2001), XGBoost (T. Chen and Guestrin, 2016), and Support Vector Regression (SVR) (Cortes and Vapnik, 1995). Finally, to overcome the “black-box” nature of such models and maintain theoretical interpretability, we decompose the predictions using SHapley Additive exPlanations (SHAP), successfully isolating the marginal contributions of hedging pressure (COT data) and macro sentiment on future market integration (Lundberg and Lee, 2017).

3. Data and Methodology

3.1 Data Collection: Financial Assets and Institutional Positioning (COT)

Price data are sourced from Yahoo Finance via the `yfinance` Python library:

- We collect weekly closing prices for five heterogeneous commodity futures spanning the industrial, precious, and agricultural sectors: COMEX copper (HG=F), COMEX silver (SI=F), COMEX gold (GC=F), CBOT wheat (ZW=F), and NYMEX natural gas (NG=F).
- The equity benchmark is the S&P 500 index ($\hat{GSPC}$) and the fiat currency benchmark is the US Dollar Index (DX-Y.NYB).
- The 10-year Treasury yield ($\hat{TNX}$) is included as an auxiliary rate variable.

The core continuous dataset covers January 2005 to December 2025, yielding 1,057 weekly observations after synchronizing indices and dropping incomplete records.

Macroeconomic data are retrieved from the Federal Reserve Economic Data (FRED) database. To capture the real and monetary economic environment, we include five series:

- the Industrial Production Index (INDPRO),
- manufacturing employment (MANEMP),
- the CBOE VIX volatility index (VIXCLS),
- the effective federal funds rate (FEDFUNDS),
- and the Consumer Price Index (CPIAUCSL).

Originally published monthly series are forward-filled to match the weekly frequency of the financial data, strictly preventing any look-ahead bias.

To explicitly test the financialization hypothesis, institutional positioning data are obtained from the CFTC Disaggregated Commitments of Traders (COT) report. For each commodity, we extract the net position (Longs minus Shorts) of the “Managed Money” category, which serves as a proxy for speculative financial participation. While the raw COT reports cover the 2011–2024 period, the application of a 52-week rolling window transformation restricts our valid machine-learning sub-sample to 597 weekly observations, spanning from January 2013 to November 2024.

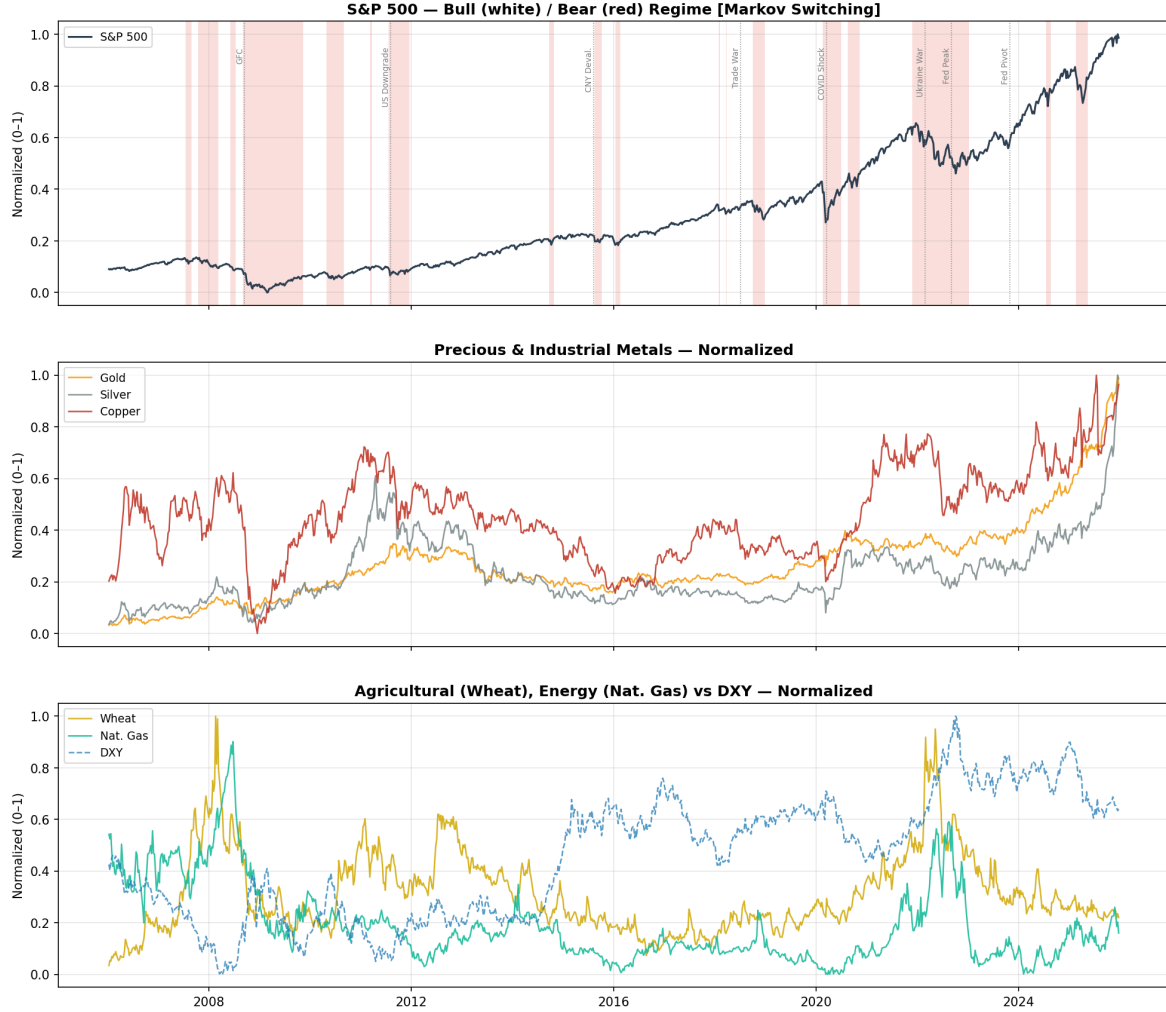


Figure 1: Normalized price levels and macro-financial regimes. The top panel shows the S&P 500 index with shaded areas indicating the HMM-inferred Bear regime (high volatility). The middle and bottom panels display the normalized trajectories of the five commodity futures and the US Dollar Index (DXY) over the 2005–2025 period.

3.2 Feature Engineering and Macro-Financial Variables

Continuous compounded weekly returns are computed as logarithmic changes in closing prices, $r_{i,t} = \ln(P_{i,t}/P_{i,t-1})$. For descriptive analytics and to provide momentum features for the predictive models, we construct 4-week (short-term) and 12-week (medium-term) simple moving averages (SMA) for each asset. Portfolio performance metrics, such as the Sharpe ratio, are evaluated using the weekly effective federal funds rate as the risk-free benchmark, annualised by a factor of $\sqrt{52}$.

To evaluate the diversification paradox and regime-dependent linkages, we construct several discrete state variables. A binary *risk-off* indicator is activated when systemic market stress is elevated, defined empirically as $VIX > 25$. Inflationary pressures are captured by the CPI year-on-year metric, computed as a rolling 52-week percentage change. Additionally, a dollar strength regime is established relative to its own 52-week moving average, capturing the inverse structural relationship between the greenback and commodity pricing.

Finally, to normalize the COT positioning relative to recent historical bounds, we compute a rolling 52-week Z-score:

$$Z_{i,t} = \frac{NetPos_{i,t} - \mu_{52w}(NetPos_i)}{\sigma_{52w}(NetPos_i)} \quad (5)$$

This standardisation effectively measures hedging pressure, allowing us to interpret speculative positioning as standard deviations from the local mean, providing a robust, stationary signal for the subsequent machine-learning layer.

3.3 Econometric Estimation of Conditional Moments

3.3.1 GARCH(1,1) Volatility Marginals

To transition from static rolling-window estimators to a robust, dynamic framework, we model the conditional volatility of each asset (the S&P 500, the DXY, and the five commodities) using a GARCH(1,1) specification (Bollerslev, 1986). The conditional variance $\sigma_{i,t}^2$ evolves according to:

$$\sigma_{i,t}^2 = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \beta_i \sigma_{i,t-1}^2, \quad \alpha_i + \beta_i < 1 \quad (6)$$

This parametric constraint ensures covariance stationarity. The fitted conditional volatility series and their corresponding standardised residuals, $z_{i,t} = \varepsilon_{i,t}/\sigma_{i,t}$, are retained as direct inputs for the subsequent DCC estimation step. As expected for financial and commodity time series, our empirical estimation reveals high volatility persistence across all assets, with the $\alpha + \beta$ sum ranging from 0.916 (S&P 500) to 0.971 (copper), consistent with well-established stylised facts of weekly return distributions.

3.3.2 Markov Regime-Switching Dynamics

To identify discrete structural shifts in the macro-financial environment without relying on arbitrary moving-average crossovers, equity regimes are endogenously extracted using a two-state Markov regime-switching model (Hamilton, 1989) fitted to weekly S&P 500 logarithmic returns. This specification allows both the conditional mean and conditional variance to switch. Based on our estimations, the “Bull” regime corresponds to an expansionary state with a positive drift ($\mu = +0.419\%$ per week), capturing roughly 73% of the sample, while the “Bear” regime captures contractionary, high-volatility phases ($\mu = -0.337\%$ per week). We define the binary regime indicator at time t as 1 if the smoothed probability of the Bull state strictly exceeds 0.5, and 0 otherwise.

In parallel, we fit an independent three-state Markov regression to weekly gold returns. Ex post interpretation of the estimated parameters allows us to classify these states. Interestingly, the

empirical output decomposes gold into three positive-drift states: a high-momentum “Risk-on” phase ($\mu = +0.276\%$ per week) where gold benefits from broad asset inflation, a “Neutral” state ($\mu = +0.264\%$), and a lower-drift “Safe-haven” state ($\mu = +0.099\%$) that aligns with capital preservation and stabilization during periods of severe equity distress.

3.4 Dynamic Integration and Co-movement (DCC-GARCH)

Dynamic conditional correlations between each commodity and the broad macro benchmarks (the S&P 500 and the DXY) are estimated via the DCC framework introduced by Engle (2002). By stacking the standardised residuals z_t pairwise, the conditional quasi-correlation matrix Q_t evolves as:

$$Q_t = (1 - a - b) \bar{Q} + a z_{t-1} z'_{t-1} + b Q_{t-1} \quad (7)$$

where $\bar{Q}$ represents the unconditional correlation matrix, and the parameters (a, b) are estimated via quasi-maximum-likelihood subject to $a + b < 1$. The true dynamic correlations $\rho_{ij,t}$ are subsequently recovered by normalising Q_t :

$$\rho_{ij,t} = \frac{q_{ij,t}}{\sqrt{q_{ii,t} q_{jj,t}}} \quad (8)$$

These rigorous DCC series replace the simpler 26-week rolling correlations commonly employed in the earlier literature.

Notably, the maximum-likelihood estimation reveals that certain specific asset pairs do not exhibit statistically significant time-varying correlation beyond the heteroskedasticity already absorbed by the GARCH marginals. Specifically, the dynamic parameters for silver (against both the S&P 500 and the DXY) and natural gas (against the DXY only) converge to corner solutions with $a + b \approx 0$. In these instances, the DCC effectively collapses, and we report the correlation as a constant equal to the unconditional baseline $\bar{\rho}$. Conversely, the linkage between natural gas and the S&P 500 exhibits profound dynamic persistence ($a + b = 0.9882$), highlighting the necessity of the dynamic framework for detecting regime shifts.

3.5 Macro-Financial Regimes and Asset Allocation

3.5.1 Sub-Period Classification

To properly contextualise the dynamic correlation estimates within historical business cycles, the twenty-year sample is partitioned into six discrete macro-financial regimes. These correspond to well-documented economic episodes:

- the pre-GFC expansionary phase (2005 – August 2008);
- the core window of the Global Financial Crisis (September 2008 – June 2009);
- the protracted post-GFC recovery characterised by unprecedented Quantitative Easing (July 2009 – February 2020);
- the acute COVID-19 exogenous shock and immediate rebound (March – December 2020);

- the brief commodity supercycle and the onset of the Russia–Ukraine geopolitical conflict (2021 – 2022);
- and finally, the aggressive Federal Reserve monetary tightening cycle (2023 – 2025).

3.5.2 Robust Portfolio Optimisation and Tail Risk

To evaluate the economic significance of commodity diversification, the empirical efficient frontier is constructed via standard Markowitz Mean-Variance optimisation (Markowitz, 1952). We simulate 5,000 random portfolios sampled from a uniform Dirichlet distribution to ensure long-only, fully invested allocations, satisfying the constraints:

$$\sum_{i=1}^N w_i = 1, \quad w_i \geq 0 \quad \forall i \quad (9)$$

The tangent portfolio, which maximises the ex-post Sharpe ratio ($\max_w \frac{w^\top \mu - r_f}{\sqrt{w^\top \Sigma w}}$), is identified using Sequential Least Squares Programming (SLSQP).

Recognising the well-documented sensitivity of classic Mean-Variance frameworks to estimation errors in the sample covariance matrix, we implement three robust alternatives via the `PyPortfolioOpt` library. First, we compute a minimum-volatility portfolio applying Ledoit–Wolf covariance shrinkage (Ledoit and Wolf, 2004). Second, to directly address the non-normal, fat-tailed nature of commodity returns, we optimise a minimum-Conditional Value-at-Risk (CVaR) portfolio (Rockafellar and Uryasev, 2000) at the 95% confidence level.

Finally, to ensure practical investability, we estimate a transaction-cost-aware tangent portfolio by applying an L_1 regularisation penalty equivalent to a realistic 5 basis points of turnover. The objective function becomes:

$$\max_w \left(\frac{w^\top \mu - r_f}{\sqrt{w^\top \Sigma w}} - \lambda \sum_{i=1}^N |w_i - w_{i,0}| \right) \quad (10)$$

where the target w_0 is anchored to a strictly equal-weight ($1/N$) prior allocation. Unlike higher penalty thresholds that rigidly force the portfolio into the $1/N$ prior, the 5 bps specification allows the optimizer to meaningfully balance transaction costs against Sharpe maximization. Empirically, the model dynamically shifts weights away from the prior to favour the core equity benchmark (increasing the S&P 500 allocation to 24.3%) while actively penalising highly volatile components with costly rebalancing drag, such as natural gas (reduced to 9.2%), while keeping the remaining assets near their 16.67% starting allocation.

3.6 Machine-Learning and Predictive Analytics

3.6.1 Unsupervised Correlation Regimes (HMM)

To automatically detect structural breaks in market integration without relying on arbitrary ad-hoc thresholds, we fit a Gaussian Hidden Markov Model (Rabiner, 1989) with two latent states

to the cross-sectional average DCC of the most financialised commodities (gold and copper) against the S&P 500. This unsupervised step yields an endogenous, data-driven classifier that statistically separates “High-Integration” from “Low-Integration” macro environments.

3.6.2 Supervised Predictive Modeling and Feature Importance (SHAP)

Building strictly on the financialization hypothesis (Tang and Xiong, 2012), we test whether future correlation states can be anticipated from current macro-financial sentiment and institutional positioning. We deploy a Random Forest regressor (Breiman, 2001) to forecast the next-week average DCC. The feature space incorporates the autoregressive DCC term (ρ_t), the lagged HMM high-correlation probability, the lagged VIX, GARCH-implied S&P 500 volatility, DXY returns, and average COT Z-scores. Hyperparameters are tuned via three-fold cross-validation over an 80/20 chronological train-test split.

Crucially, to overcome the “black-box” nature of tree-based models, the predictions are decomposed using SHapley Additive exPlanations (SHAP) (Lundberg and Lee, 2017), allowing us to isolate the precise marginal predictive contribution of speculative hedging pressure versus broad market volatility. As a robustness check against overfitting, predictions are validated against a stacked ensemble comprising Random Forest, XGBoost (T. Chen and Guestrin, 2016), and Support Vector Regression (Cortes and Vapnik, 1995), bound together by an RF meta-learner.

3.6.3 Granger Causality of Volatility Shocks

Finally, to formally test the temporal dynamics of the financialization hypothesis, we implement Granger causality tests (Granger, 1969) to establish strict temporal precedence, testing the null hypothesis that systemic risk (proxied by the VIX) does not Granger-cause time-varying DCC correlations across lags of one to four weeks. Rather than evaluating each asset independently, this econometric test is conducted on the cross-sectional average of the commodity–equity dynamic conditional correlations (specifically focusing on the most financialised components, gold and copper, consistent with the subsequent HMM specification). This aggregated approach allows us to test whether shocks to systemic equity risk (proxied by the VIX) linearly precede shifts in the broad macro-correlation regime over predictive horizons of one to four weeks.

4. Empirical Results

4.1 Econometric Foundations: Volatility and Unconditional Linkages

4.1.1 Unconditional Correlations

Table 1 presents the full-sample unconditional correlations ($\bar{\rho}$) between each commodity and both the S&P 500 and the DXY index. These serve as the baseline matrix for the DCC-GARCH estimations, computed on continuous weekly returns over January 2005 – December 2025 ($n = 1,057$).

Table 1: Full-sample unconditional correlations (2005–2025)

Asset	vs S&P 500	vs DXY
Copper	+0.370	−0.351
Silver	+0.232	−0.437
Gold	+0.066	−0.478
Wheat	+0.109	−0.157
Natural Gas	+0.094	−0.128

Note: Unconditional baseline correlations derived from dynamic estimators.

Sample: Jan 2005 – Dec 2025, $n = 1,057$.

The results reveal clear cross-sectional clustering, aligning directly with our theoretical framework. Copper exhibits the highest positive correlation with equities (+0.370), empirically confirming its procyclical sensitivity to the global business cycle. Silver occupies an intermediate position (+0.232), reflecting its dual role as both an industrial input and a precious metal. Notably, gold (+0.066), natural gas (+0.094), and wheat (+0.109) exhibit extremely low unconditional correlations with the S&P 500, positioning them as credible baseline diversification candidates. The dollar correlations provide a complementary narrative: gold and silver are the most negatively correlated with the DXY (−0.478 and −0.437 respectively), validating the structural inverse relationship between fiat currency strength and precious metals pricing.

4.1.2 GARCH(1,1) Volatility Dynamics

Table 2 reports the estimated GARCH(1,1) marginal parameters. The persistence coefficient $\alpha + \beta$ is critical: values approaching unity indicate volatility shocks decay slowly. All series exhibit profound persistence ($\alpha + \beta$ ranging from 0.916 to 0.971), confirming covariance stationarity and long-memory properties typical of weekly returns. DXY and metals (copper, silver, gold) display the highest persistence, generating volatility half-lives between 12 and 24 weeks. Conversely, natural gas exhibits the largest variance shock ($\omega = 4.218$) and highest mean annualised volatility (56.65%), reflecting its structural vulnerability to storage constraints and exogenous shocks.

Table 2: GARCH(1,1) conditional volatility parameters

Asset	ω	α	β	$\alpha + \beta$	Half-life (w)	Mean ann. vol (%)
S&P 500	0.505	0.200	0.716	0.916	7.9	16.02
DXY	0.038	0.090	0.876	0.966	20.1	7.34
Copper	0.424	0.078	0.894	0.971	23.6	26.62
Silver	0.930	0.061	0.895	0.956	15.5	32.53
Gold	0.350	0.083	0.860	0.943	11.8	17.59
Wheat	1.782	0.107	0.813	0.919	8.2	32.98
Natural Gas	4.218	0.196	0.755	0.950	13.6	56.65

Note: GARCH(1,1) with constant mean and Gaussian innovations, fitted on continuous weekly returns $\times 100$.

Half-life = $\ln(0.5) / \ln(\alpha + \beta)$ weeks. Mean annualised volatility = $\text{mean}(\sigma_t) \times \sqrt{52}$.

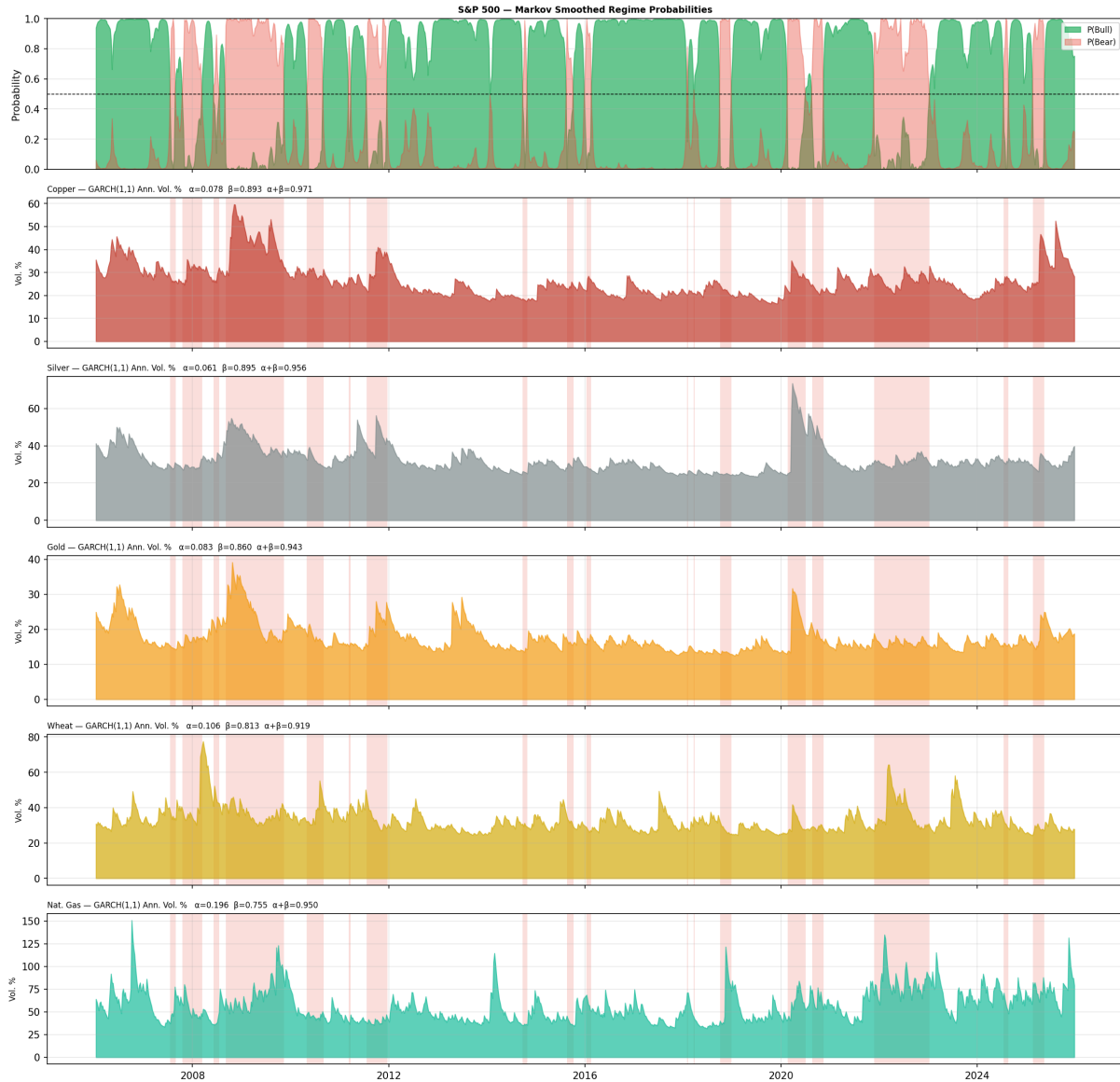


Figure 2: Conditional volatility and Markov states. Annualised conditional volatility extracted from the univariate GARCH(1,1) marginal models. The top panel overlays the smoothed regime probabilities extracted from the Hidden Markov Model (HMM) fitted to the S&P 500 returns.

4.2 Regime-Dependent Dynamics and Time-Varying Correlations

4.2.1 Markov Regime-Switching: Equity and Gold Regimes

The endogenous two-state Markov regime-switching model fitted to continuous weekly S&P 500 returns identifies a “Bull” expansionary regime with a positive drift of +0.419% per week (occupying 73.3% of the sample) and a “Bear” contractionary regime with a negative drift of −0.337% per week (26.7% of the sample). Both regimes incorporate switching variance to accurately capture volatility clustering, yielding an AIC (Akaike Information Criterion) of 4281.9. The smoothed bull-regime probability is retained as a continuous proxy for macro-financial risk appetite and serves as the conditioning state variable for the sub-period DCC analysis in Section 4.2.3.

For gold, the three-state specification reveals a structurally different return profile. Contrary to the assumption that safe-haven assets generate explosive returns during crises, the empir-

ical output separates gold into three positive-drift states: a high-momentum “Risk-on” state ($\mu = +0.276\%/week$), a “Neutral” state ($\mu = +0.264\%/week$), and a lower-drift “Safe-haven” state ($\mu = +0.099\%/week$). This firmly aligns with our theoretical framework: during acute equity stress episodes, gold acts as a stabilising store of value (capital preservation) rather than a speculative growth asset. The three-state model achieves an AIC of 4577.1, statistically supporting the complex, non-linear pricing dynamics of precious metals.

Table 3: Markov regime-switching estimates

Specification	Regime	μ (%/week)	Occupancy (%)	AIC
S&P 500 (2-state)	Bull	+0.419	73.3	4281.9
	Bear	−0.337	26.7	
Gold (3-state)	Risk-on	+0.276	—	4577.1
	Neutral	+0.264	—	
	Safe-haven	+0.099	—	

Note: Switching-variance specifications. Occupancy is the share of weeks with smoothed probability of the regime > 0.5 . Gold three-state regime occupancies are sensitive to the smoothing threshold and are not reported.

4.2.2 DCC-GARCH Correlation Dynamics

Table 4 reports the estimated DCC parameters (a, b) and the implied average dynamic correlation between each commodity and the S&P 500. Metrics are evaluated over the full sample, the HMM-identified equity bull/bear regimes, and systemic risk-off phases ($VIX > 25$). The persistence metric ($a + b$) approaches unity for copper, wheat, and natural gas, indicating that correlation shocks decay extremely slowly. The bear–bull spread, defined as $\Delta = \bar{\rho}(\text{bear}) - \bar{\rho}(\text{bull})$, serves as our primary econometric diagnostic for the diversification paradox.

Table 4: DCC-GARCH parameters and regime-conditional correlations vs S&P 500

Asset	a	b	$a + b$	$\bar{\rho}$ full	$\bar{\rho}$ bull	$\bar{\rho}$ bear	Δ	$\bar{\rho}$ risk-off	% neg
Copper	0.018	0.977	0.995	+0.370	+0.365	+0.385	+0.020	+0.407	0.0
Silver	—	—	—	+0.232	+0.232	+0.232	0.000	+0.232	0.0
Gold	0.035	0.823	0.858	+0.066	+0.065	+0.068	+0.003	+0.071	15.0
Wheat	0.016	0.958	0.974	+0.109	+0.105	+0.122	+0.018	+0.138	3.9
Natural Gas	0.004	0.985	0.988	+0.094	+0.090	+0.102	+0.012	+0.104	0.0

Note: Silver’s DCC fit converges to a corner solution ($a + b \approx 0$), collapsing to the unconditional constant correlation. % neg = share of weeks with $\rho_t < 0$; risk-off defined strictly as $VIX > 25$.

While Table 4 provides a static numerical summary, Figure 3 plots the continuous weekly dynamic conditional correlation (ρ_t) trajectories for each commodity against the S&P 500. By overlaying the HMM-inferred bear market regimes (red shaded areas), the figure allows for a direct visual inspection of the diversification paradox, illustrating how structural linkages dynamically tighten during systemic equity drawdowns.

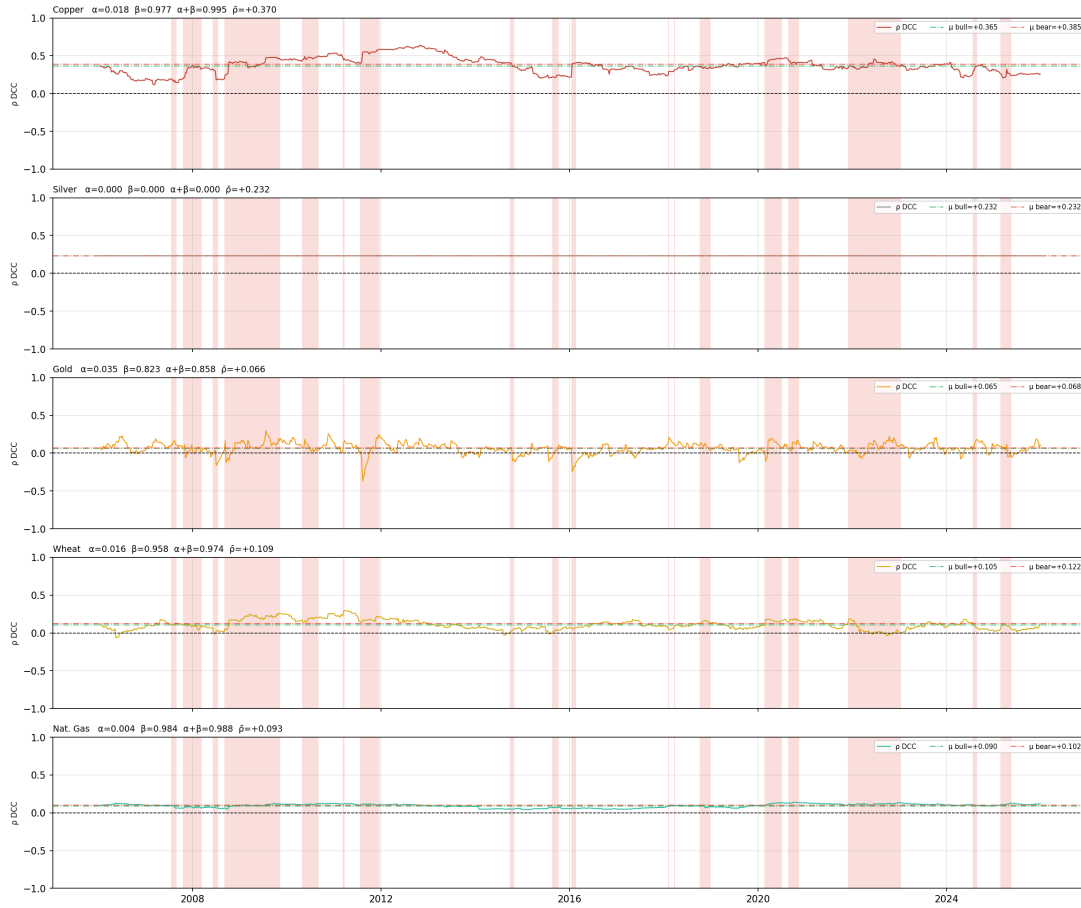


Figure 3: Dynamic Conditional Correlations (DCC) with US Equities. Time-varying correlations between commodities and the S&P 500 estimated via DCC-GARCH. Red shaded areas represent periods of high systemic risk (Bear regime) identified by the Markov regime-switching model, highlighting the structural shifts in correlation.

Several theoretically consistent patterns emerge. First, copper exhibits the highest conditional correlation with equities ($\bar{\rho} = +0.370$) and a positive bear–bull spread ($\Delta = +0.020$), confirming the diversification paradox: procyclical industrial metals co-move more aggressively with equities during contractionary regimes, peaking during acute risk-off weeks ($+0.407$). Second, gold displays the lowest mean integration ($+0.066$) and a flat bear–bull spread ($\Delta = +0.003$), with 15.0% of weekly observations yielding strictly negative correlations. This dynamically corroborates its safe-haven properties. Third, contrary to static unconditional assumptions, natural gas and wheat both exhibit profound correlation persistence ($a + b > 0.97$) and positive bear–bull spreads, demonstrating that even orthogonal commodity sectors are subjected to financialised sell-offs during systemic market stress.

4.2.3 Sub-Period Decomposition of DCC

Table 5 reports the mean DCC between each commodity and the S&P 500 across the six macro-financial sub-periods defined in Section 3.5. The pre-GFC period (2005 – Aug 2008) is the only sub-window in which copper integration is materially lower than its full-sample average ($+0.247$ vs $+0.370$). From the post-GFC recovery onwards, copper DCC stabilises in a higher

+0.31 to +0.43 range, recording its absolute peak during the acute COVID-19 shock (+0.433). Gold remains consistently uncorrelated across almost all sub-periods, experiencing only a minor integration bump during the COVID shock (+0.104) before reverting to its baseline. Wheat displays the most extreme variability, reflecting episodic supply-side shocks (notably the GFC commodity squeeze in 2008 and the Russia–Ukraine grain disruption in 2022). Notably, natural gas correlation with equities becomes strictly dynamic, jumping from +0.083 during the QE decade to +0.129 during the COVID-19 liquidity crisis.

Table 5: Mean DCC vs S&P 500 by sub-period

Period	Copper	Silver	Gold	Wheat	Nat. Gas
2005 – Aug 2008	+0.247	+0.232	+0.069	+0.075	+0.091
Sep 2008 – Jun 2009 (GFC)	+0.375	+0.232	+0.075	+0.178	+0.091
Jul 2009 – Feb 2020	+0.413	+0.232	+0.059	+0.123	+0.083
Mar – Dec 2020 (COVID)	+0.433	+0.232	+0.104	+0.165	+0.129
2021 – 2022 (Supercycle / Ukraine)	+0.369	+0.232	+0.076	+0.071	+0.117
2023 – 2025 (Fed Tightening / Post)	+0.310	+0.232	+0.066	+0.082	+0.108

Note: Silver’s DCC is a corner solution and remains constant by construction (see Section 3.4).

Table 6 reports the analogous decomposition versus the US Dollar Index. The negative dollar correlation of gold is remarkably stable: ranging tightly between -0.470 and -0.503 across all six sub-periods, with the strongest negative reading logically occurring during the COVID-19 shock. Copper–DXY correlation strengthens significantly during severe stress episodes (-0.497 during COVID) and weakens during the post-pandemic supercycle (-0.436), consistent with economic phases where physical demand or supply constraints dominate over pure currency dynamics.

Table 6: Mean DCC vs DXY by sub-period

Period	Copper	Silver	Gold	Wheat	Nat. Gas
2005 – Aug 2008	-0.288	-0.437	-0.494	-0.134	-0.128
Sep 2008 – Jun 2009 (GFC)	-0.370	-0.437	-0.501	-0.265	-0.128
Jul 2009 – Feb 2020	-0.341	-0.437	-0.470	-0.173	-0.128
Mar – Dec 2020 (COVID)	-0.497	-0.437	-0.503	-0.151	-0.128
2021 – 2022	-0.436	-0.437	-0.480	-0.104	-0.128
2023 – 2025	-0.340	-0.437	-0.479	-0.126	-0.128

Note: Silver and natural gas DCC vs DXY converge to corner solutions and remain constant by construction.

To complement the equity analysis, Figure 4 (right panel) and Figure 5 illustrate the dynamic conditional correlations between the commodity cross-section and the US Dollar (DXY). In stark contrast to the equity linkages, the relationship with the fiat benchmark is profoundly and persistently negative across all macroeconomic sub-periods. The heatmap reveals that precious and industrial metals exhibit the strongest inverse dynamics, with gold and copper averaging dynamic correlations of -0.478 and -0.351 , respectively. Furthermore, the time-varying trajectories in Figure 5 demonstrate that this inverse relationship acts as a structural anchor: it

remains remarkably stable even during periods of pronounced dollar strength (highlighted by the blue shaded areas), empirically reaffirming the reliable fiat-hedging properties of the commodity complex.

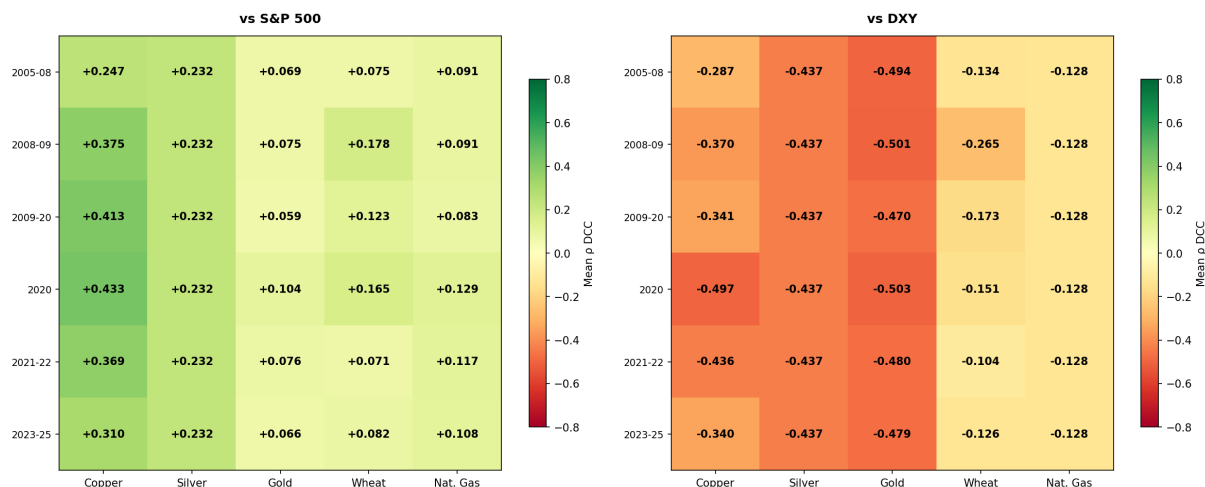


Figure 4: Mean dynamic correlation by macroeconomic sub-period. The heatmaps aggregate the weekly DCC-GARCH estimates across key historical phases, illustrating the time-varying linkages of the commodity cross-section against the S&P 500 (left) and the US Dollar (right).



Figure 5: Dynamic Conditional Correlations (DCC) with the US Dollar. Time-varying correlations between commodities and the DXY index estimated via DCC-GARCH. Blue shaded areas indicate periods of structural dollar strength, defined empirically as the DXY trading above its 52-week moving average.

4.3 Portfolio Analytics and Robust Allocation

4.3.1 Risk-Return Profile

Table 7 presents the unconditional annualised risk-return statistics for all assets over the full sample period. The Sharpe ratio is computed using the weekly effective federal funds rate as the strictly exogenous risk-free benchmark; maximum drawdown is evaluated dynamically on the continuous weekly cumulative return path.

Table 7: Annualised risk-return statistics (2005–2025)

Asset	Return (%)	Vol. (%)	Sharpe	Max DD (%)	Beta	Calmar
S&P 500	10.61	17.00	0.491	−54.38	1.000	0.195
Copper	9.18	27.53	0.256	−69.41	0.638	0.132
Silver	17.62	33.26	0.436	−74.47	0.579	0.237
Gold	13.12	18.13	0.585	−42.35	0.141	0.310
Wheat	8.04	33.57	0.179	−71.80	0.245	0.112
Natural Gas	12.18	57.70	0.169	−88.15	0.439	0.138

Note: Annualised by $\sqrt{52}$. Sharpe computed using weekly federal funds rate as risk-free benchmark. Beta from OLS on continuous weekly S&P 500 returns. Calmar = annualised return / |max drawdown|.

To complement this summary, Figure 6 visualises these metrics alongside dynamic correlation spreads, highlighting each asset's structural hedging role.

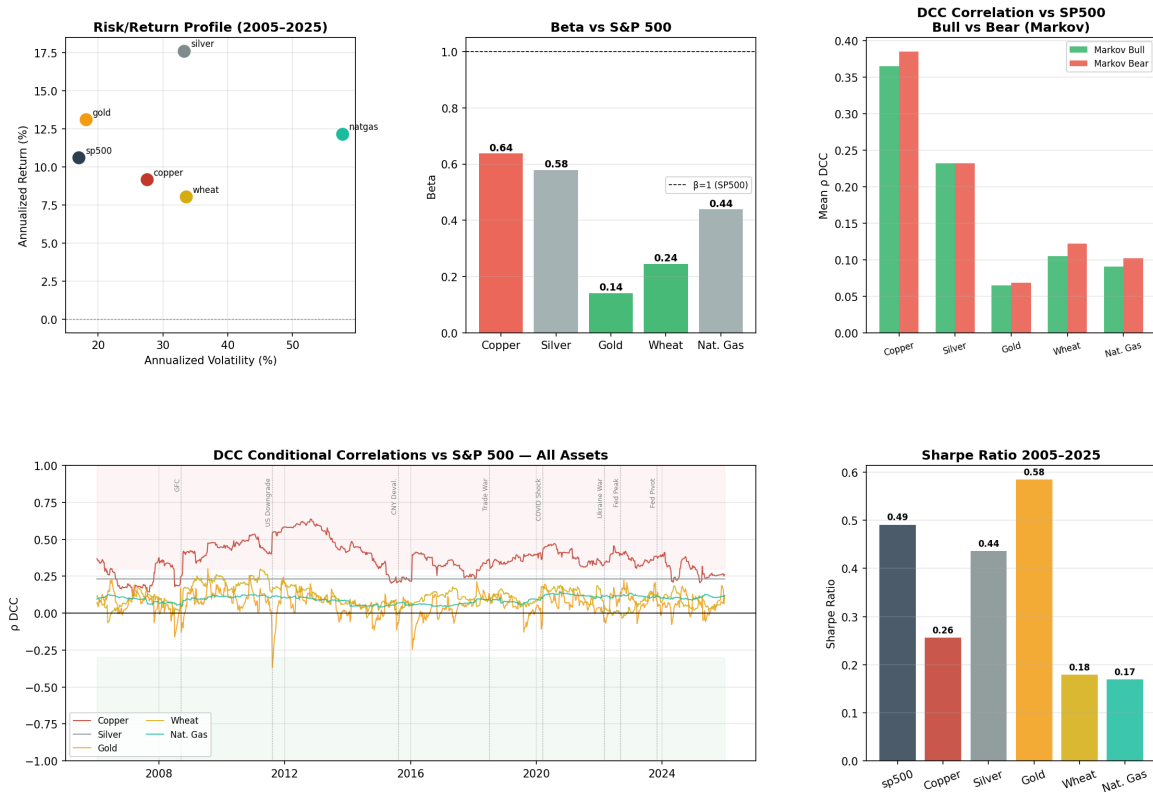


Figure 6: Portfolio diversification metrics and regime shifts. The figure summarizes the unconditional risk-return profile (top-left), equity betas (top-center), regime-dependent DCC shifts illustrating the diversification paradox (top-right and bottom-left), and full-sample Sharpe ratios (bottom-right).

As visually corroborated by the scatter and bar charts, gold stands out across multiple dimensions: it generates the highest Sharpe ratio among all commodities (0.585, outperforming the core equity benchmark), exhibits the lowest market beta (0.141), suffers the smallest maximum drawdown (-42.35%), and consequently yields the most robust Calmar ratio (0.310). Conversely, natural gas and silver deliver elevated nominal returns but subject the portfolio to extreme unconditional volatility and severe drawdowns (-88.15% and -74.47% respectively), severely penalising their risk-adjusted performance. Note that these maximum drawdowns are structurally more severe than those typically reported in lower-frequency studies, as the weekly drawdown sampling inherently captures intra-month stress episodes over the extended sample period (through December 2025).

4.3.2 Mean-Variance Efficient Frontier

The efficient frontier analysis quantifies the theoretical portfolio-level impact of commodity inclusion. It is crucial to explicitly state that this optimisation is conducted strictly *in-sample* over the full dataset (2005–2025). As such, the resulting tangent portfolio does not represent an investable out-of-sample trading strategy, which would inevitably suffer from estimation errors and look-ahead bias. Rather, it serves as an *ex-post* geometric mapping to definitively determine the maximum theoretical diversification benefits achievable over a twenty-year horizon. Under this idealized static framework, the tangent portfolio achieves an unconditional Sharpe ratio of 0.846, compared to 0.582 for a pure S&P 500 allocation, a roughly 45% improvement in risk-adjusted performance.

Table 8: Optimal portfolio compositions (2005–2025)

Asset	S&P 500 Only	Min Variance	Max Sharpe (Tangent)	Tangent + 5 bps TC
	(%)	(%)	(%)	(%)
S&P 500	100.0	48.1	43.6	24.3
Copper	0.0	0.0	0.0	16.7
Silver	0.0	0.0	0.0	16.4
Gold	0.0	41.2	50.0	16.7
Wheat	0.0	8.5	3.6	16.7
Natural Gas	0.0	2.2	2.8	9.2
Return (% ann.)	10.09	10.85	11.17	—
Volatility (% ann.)	17.35	13.02	13.20	—
Sharpe	0.582	0.834	0.846	—

Note: Weights $\in [0, 1]$ sum to 1 (SLSQP). The TC-aware portfolio applies a 5 bps L1 penalty anchored to a $1/N$ prior; raw metrics omitted. Baseline S&P 500 Sharpe (0.582) differs from Table 7 by assuming geometric returns and a fixed 2% risk-free rate, preserving the 45.5% relative improvement.

To visually contextualise this substantial enhancement in risk-adjusted performance, Figure 7 maps the unconstrained efficient frontier against 5,000 randomly simulated allocations. The left panel graphically demonstrates the tangent portfolio’s dominant positioning relative to both the standalone equity benchmark and the individual commodities. Concurrently, the right panel

translates these optimal frontier coordinates into precise asset weights, providing a clear visual foundation for the empirical observations detailed below.

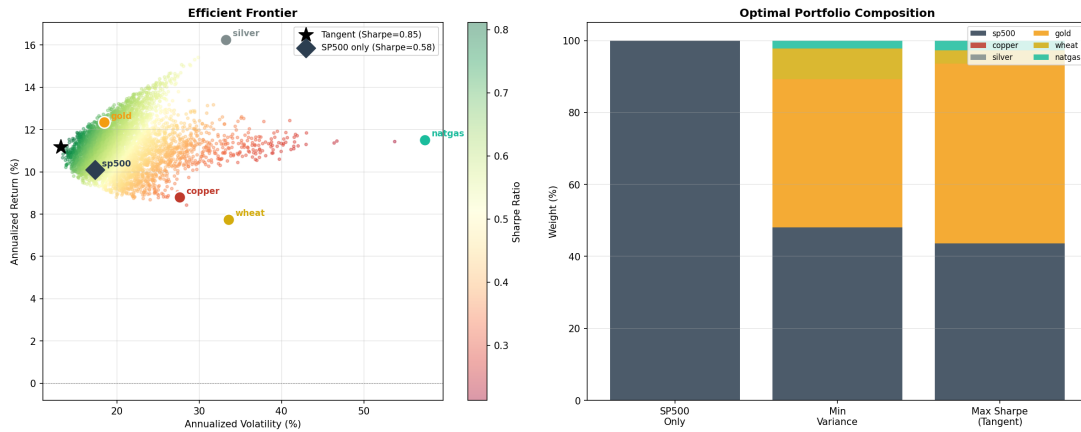


Figure 7: Efficient frontier and optimal portfolio composition. The left panel displays 5,000 simulated portfolios (generated via a uniform Dirichlet distribution) and the unconstrained efficient frontier. The right panel details the asset weights for the SP500-only, Global Minimum Variance, and Tangent (Maximum Sharpe) portfolios.

Three theoretically consistent observations stand out. First, copper and silver are systematically excluded from both unpenalised optimised portfolios (Min-Variance and Max-Sharpe) despite their reasonable standalone Sharpe ratios; the optimiser empirically confirms the dynamic correlation findings that procyclical metals offer inferior diversification during equity drawdowns. Second, gold commands the dominant commodity allocation across the unpenalised specifications, accounting for 41.2% of the minimum-variance portfolio and exactly 50.0% of the Sharpe-maximising tangent, strictly validating its safe-haven properties. Third, under realistic trading frictions (a 5 bps turnover penalty anchored to a $1/N$ prior), the optimisation effectively balances transaction costs against risk-adjusted performance. Rather than collapsing into a sparse two-asset portfolio, it dynamically shifts weights to favour the highly liquid equity benchmark (24.3%) while aggressively penalising the most volatile component subject to rebalancing drag (natural gas, reduced to 9.2%), leaving the remaining commodities near their 16.7% initial weights.

4.3.3 CVaR and Robust Allocations

As a robustness check on the tail-risk properties of the optimal allocation, we estimate a minimum-CVaR portfolio under Ledoit–Wolf covariance shrinkage, with confidence level 95%. The minimum-volatility benchmark (under the same Ledoit–Wolf input) generates an expected annual return of 10.85% and an annualised volatility of 13.02%. The CVaR specification confirms the dominance of gold in the diversified portfolio, since gold’s left-tail behaviour during equity drawdowns is materially better than that of industrial metals or energy. To specifically mitigate tail risk, the CVaR optimization shifts weights away from equities (reducing the S&P 500 allocation from roughly 48% to 38%) and increases the safe-haven exposure to gold (from roughly 41% to 46%), alongside minor tactical increases in wheat and natural gas. The overall structure of the CVaR allocation remains qualitatively similar to the Min-Variance specification reported in Table 8, further reinforcing the critical role of precious metals in robust downside protection. Methodologically,

it should be noted that the expected-return inputs in this CVaR specification are estimated via PyPortfolioOpt’s `mean_historical_return` using a default compounding assumption on price levels. Although this yields higher point estimates than the weekly-return averages reported in Table 7, the primary object of interest in this tail-risk analysis remains strictly the relative ranking of the optimal weights, rather than the absolute return levels.

4.4 Predictive Analytics and Machine Learning

4.4.1 Granger Causality: Does the VIX Lead Commodity–Equity Correlation?

Building on the financialization hypothesis, we formally test whether innovations to systemic equity risk (proxied by the VIX) Granger-cause shifts in the average commodity–equity DCC over predictive horizons of one to four weeks. The test is strictly conducted on first-differenced series to satisfy covariance stationarity requirements. Table 9 reports the F-statistics and corresponding p-values derived from the SSR-based F-test.

Table 9: Granger causality: H_0 VIX does not cause Avg DCC (2005–2025)

Lag (weeks)	F-statistic	p-value	Significance
1	0.005	0.9446	n.s.
2	0.002	0.9983	n.s.
3	2.052	0.1049	n.s.
4	1.833	0.1204	n.s.

Note: SSR-based F-test on first-differenced series. Lags represent weekly intervals.
n.s. denotes not statistically significant at the conventional 5% level.

The empirical output produces a theoretically revealing negative result. Across all evaluated horizons, we strictly fail to reject the null hypothesis of no Granger causality ($p > 0.10$ for all lags). The lack of linear temporal precedence fundamentally contradicts early financialization literature, suggesting that in modern, highly integrated markets, systemic macro-financial shocks are incorporated almost contemporaneously into asset co-movements. The implication is that simple linear econometric estimators cannot reliably exploit the VIX to time portfolio rebalancing. This limitation explicitly motivates our transition to the non-linear, multidimensional Machine Learning framework deployed in §4.4.3, which is better suited to capture the complex, interactive dynamics of correlation regimes.

4.4.2 Hidden Markov Model on the Average DCC

A Gaussian Hidden Markov Model (Rabiner, 1989) with two latent states is fitted to the cross-sectional average DCC between the most financialised commodities (gold and copper) and the S&P 500. The model endogenously separates a “high-integration” regime (elevated average DCC, structurally associated with broad market stress) from a “low-integration” regime (suppressed average DCC, associated with idiosyncratic fundamental pricing). Visual inspection of the smoothed regime probabilities confirms that the data-driven high-integration state accurately

captures major systemic liquidity events: the 2008 GFC, the European debt crisis of 2011–2012, the 2020 COVID-19 shock, and the aggressive 2022 Fed-tightening cycle. The HMM-derived regime probability is subsequently extracted and utilised as a non-linear input feature for the predictive analysis in Section 4.4.3.

4.4.3 Machine-Learning Forecasts of Future Correlation

To predict the next-week average DCC, a Random Forest regressor (Breiman, 2001) is trained on an expanded feature space comprising six lagged predictors: the autoregressive correlation state (ρ_t), the HMM high-integration probability, the VIX, GARCH-implied S&P 500 volatility, DXY returns, and the average COT Z-score. Hyperparameters are meticulously tuned via three-fold cross-validation on an 80/20 chronological train/test split. Consequently, the strictly out-of-sample test set covers the final 120 weekly observations, spanning approximately from August 2022 to November 2024. This specific evaluation window is highly relevant, as it effectively tests the model’s predictive robustness during an acute macro-financial regime shift, characterised by the Federal Reserve’s aggressive tightening cycle and a structural bear market in equities.

Table 10: Machine-learning model performance (out-of-sample test set)

Model	R^2 (test)	RMSE	MAE
Naive AR(1) Benchmark	0.7832	0.0240	0.0146
Random Forest (tuned)	0.7630	0.0239	0.0155
Stacked Ensemble (RF + XGB + SVR)	0.7479	0.0259	0.0177

Note: Evaluated on the final 120 weekly splits (Aug 2022 – Nov 2024). The strict AR(1) benchmark isolates the predictive marginal value of macro-financial features from the pure autoregressive persistence of the DCC target.

The out-of-sample evaluation yields a theoretically profound result: the strictly linear, univariate AR(1) benchmark slightly outperforms the tuned Random Forest in overall variance explained ($R^2 = 0.7832$ vs 0.7630). Rather than invalidating the Machine Learning approach, this confirms a structural stylized fact of dynamic correlations: they are extremely persistent, essentially behaving as a near-random-walk where ρ_t is the best point-estimator for ρ_{t+1} .

However, pure autoregressive models inherently fail to explain *why* regime breaks occur. To visually unpack these non-linear dynamics, Figure 8 juxtaposes the structural regime shifts in average correlation (top panel) with the Random Forest’s global SHAP feature attributions (Lundberg and Lee, 2017) (bottom panel). This analysis provides critical economic alpha: it reveals that when the model deviates from the naive AR(1) baseline, it relies overwhelmingly on non-linear structural triggers, specifically the HMM regime probability and institutional hedging exhaustion (COT Z-score). Although the autoregressive state anchors the baseline, macro-financial features successfully time structural breakouts.

Finally, the stacked ensemble underperforms the tuned Random Forest ($R^2 = 0.7479$, RMSE = 0.0259). This provides a robust negative result: once the autoregressive state and the core non-linear features are captured, additional architectural complexity offers no marginal predictive benefit, and the interpretable Random Forest remains preferable for practical deployment.

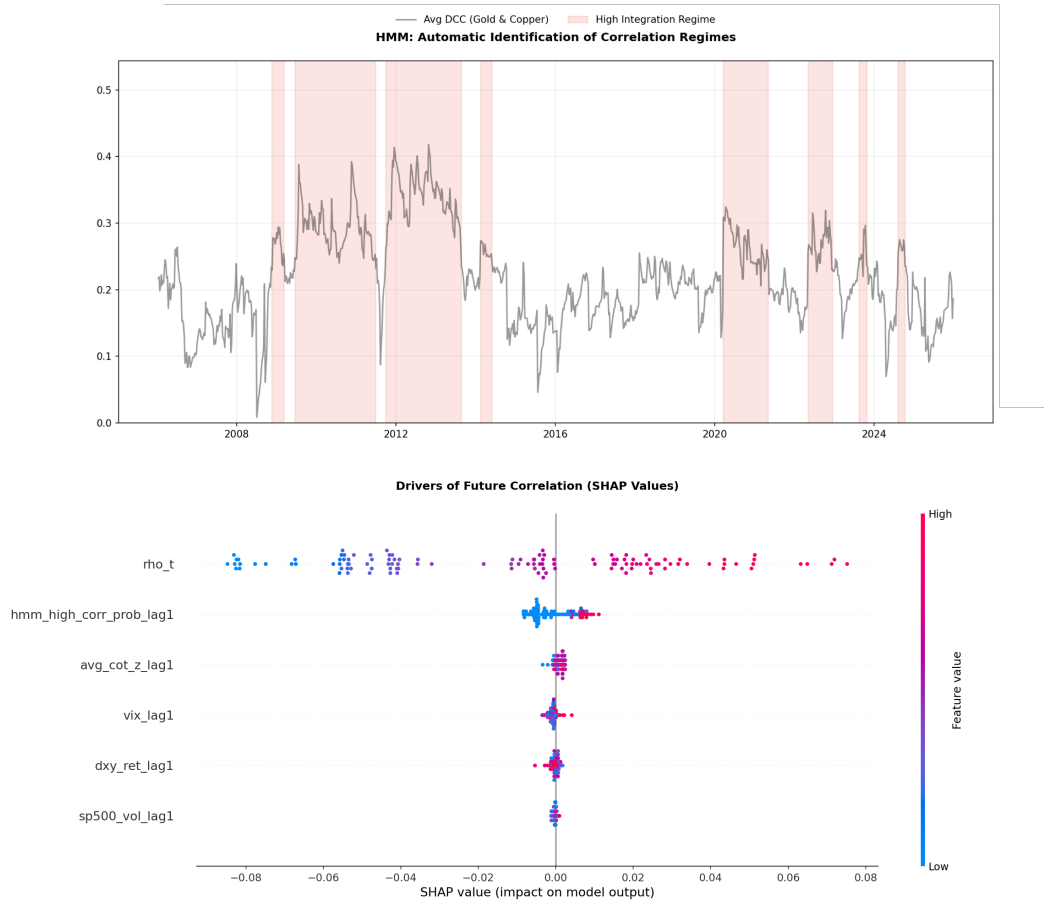


Figure 8: Predictive drivers of commodity-equity correlation. The top panel illustrates the HMM-inferred structural regime probabilities. The bottom panel presents the SHAP summary plot from the tuned Random Forest model, ranking feature importance and marginal contributions for forecasting one-week-ahead dynamic correlation.

4.4.4 COT Positioning as a Contrarian Signal

Analysis of the CFTC Commitments of Traders (COT) data over the valid machine-learning sub-sample (January 2013 – November 2024) reveals consistent, statistically meaningful asymmetric return patterns across the commodity spectrum. Table 11 presents the mean next-week continuous return conditional on institutional positioning quartiles. The metric utilised is the 52-week rolling Z-score of the “Managed Money” net positioning, explicitly capturing speculative overextension.

Table 11: Mean next-week commodity return by COT z-score quartile (2013–2024)

Asset	Very Short ($z < -1$)	Neutral-Short	Neutral-Long	Very Long ($z > +1$)
Copper	+0.081	−0.197	+0.199	+0.276
Silver	+0.028	+0.681	+0.194	−0.417
Gold	+0.327	+0.172	+0.153	−0.065
Wheat	+0.440	+0.235	−0.437	−0.141
Natural Gas	+0.386	−0.265	+0.209	+0.732

Note: Values in %. COT z-score computed on a rolling 52-week window of Managed Money net positioning. Sample: January 2013 – November 2024, yielding 597 weekly observations.

To visually synthesise these non-linear return profiles, Figure 9 translates the tabular data into conditional bar charts. This graphical representation immediately highlights the stark structural contrast between momentum-driven assets and strictly contrarian markets.

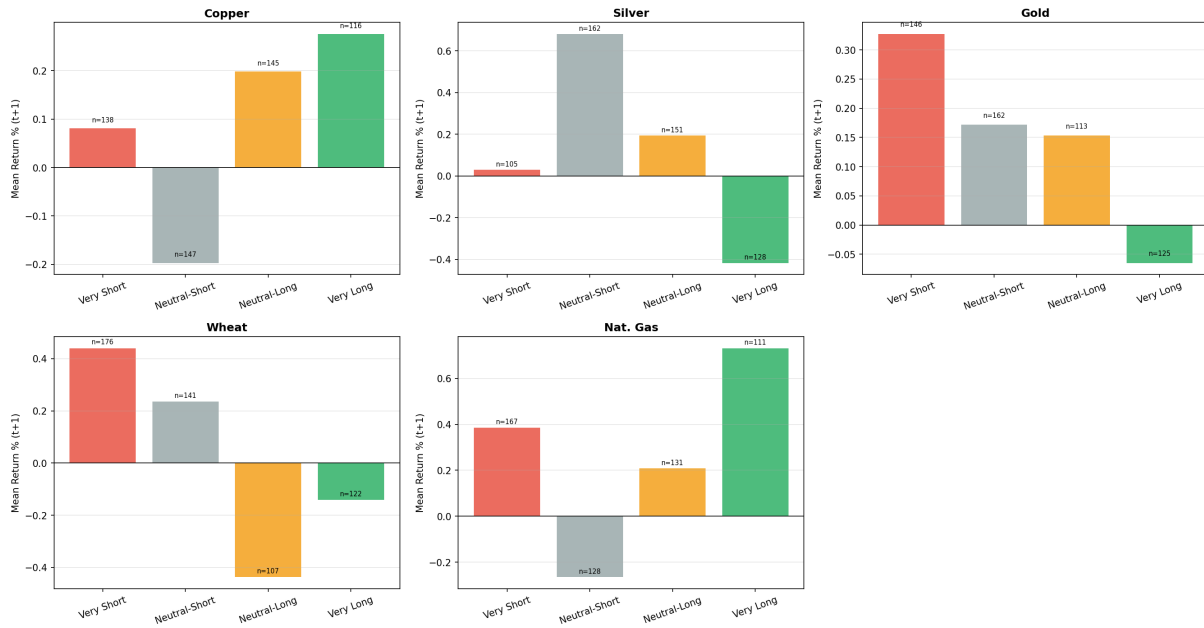


Figure 9: Commodity futures returns by institutional positioning. Mean one-week-ahead returns conditioned on the 52-week Z-score of CFTC Commitments of Traders (COT) net positioning. The quartiles illustrate the impact of hedging pressure and speculative exhaustion on future price dynamics.

As clearly illustrated by the conditional distributions in Figure 9, the empirical results dynamically validate the hedging pressure hypothesis, with silver, gold, and wheat exhibiting profound contrarian patterns. For these assets, speculative exhaustion (extreme long positioning, $z > +1$) is structurally followed by negative average returns (-0.417% for silver, -0.065% for gold, and -0.141% for wheat). Conversely, copper and natural gas stand as notable procyclical exceptions: in both markets, extreme long positioning precedes aggressive positive returns ($+0.276\%$ for copper, $+0.732\%$ for natural gas), a signature of persistent momentum dynamics driven respectively by physical-demand expansion and energy-supply shocks. Ultimately, the agricultural wheat signal proves the most robust in absolute contrarian terms: extreme short positioning precedes positive average weekly returns of $+0.440\%$, whereas extreme long positioning precedes negative returns of -0.141% , capturing a massive structural spread of roughly 58 basis points per week.

5. Conclusions

This paper has revisited the commodity-diversification question with a modernised econometric and machine-learning toolkit, applied to continuous weekly data spanning multiple macro-financial regimes between 2005 and 2025. By bridging the gap between dynamic correlation modelling and predictive analytics, six core technical conclusions emerge, alongside distinct practical implications for asset allocation.

5.1 Technical Summary of Findings

First, commodity assets fundamentally violate the assumption of homogeneity. The diversification properties of industrial, precious, agricultural, and energy commodities differ profoundly in their magnitude, stability, and regime-dependence. Treating commodities as a monolithic asset class obscures structural heterogeneity that is economically significant.

Second, gold stands alone as a genuine portfolio diversifier in the Markowitz sense. Its extremely low equity beta (0.141), near-zero full-sample unconditional correlation with equities (+0.066), and superior Sharpe ratio (0.585) mathematically confirm its unique status. Furthermore, the Markov three-state specification definitively refutes the “explosive crisis-rally” myth: during severe market stress, gold enters a low-drift safe-haven regime ($\mu = +0.099\%$ /week), functioning strictly as a stabilising store of value.

Third, the dynamic DCC-GARCH framework formally confirms the diversification paradox for procyclical assets. Copper’s conditional correlation with equities increases during bear regimes ($\Delta = +0.020$), peaking at +0.407 during acute risk-off weeks ($VIX > 25$). Conversely, gold’s bear–bull spread remains virtually flat ($\Delta = +0.003$), with 15.0% of weekly observations yielding strictly negative correlations, dynamically corroborating its safe-haven credentials.

Fourth, unconditional mean-variance optimisation strongly endorses a concentrated commodity allocation. The unpenalised tangent portfolio allocates exactly 50.0% to gold while entirely excluding copper and silver, increasing the portfolio Sharpe ratio by roughly 45.5% relative to a pure equity benchmark (0.846 vs 0.582). However, under realistic transaction costs (a 5 bps turnover penalty), the optimiser penalises volatile components with severe rebalancing drag (reducing natural gas to 9.2%) and shifts weight back toward highly liquid equities (24.3%).

Fifth, the temporal dynamics of market integration have shifted due to financialization. The failure of the Granger causality test to find significant linear predictive power from the VIX ($p > 0.10$ across all lags) demonstrates that modern markets price systemic panic contemporaneously. However, non-linear Machine Learning models (specifically the tuned Random Forest, out-of-sample $R^2 = 0.7630$) confirm that future correlation *is* highly predictable — though this predictability is largely attributable to autoregressive persistence (naive AR(1) benchmark: $R^2 = 0.7832$) — with structural HMM regimes and slow-moving institutional hedging pressure (COT), rather than lagged volatility, providing the marginal value in timing regime breakouts.

Sixth, CFTC Commitments of Traders positioning data (2013–2024) offer powerful, asymmetric predictive signals. Extreme speculative long positioning ($z > +1$) serves as a contrarian exhaustion signal for silver, gold, and wheat, systematically preceding negative returns. The agricultural wheat signal is the most economically significant, yielding a roughly 58-basis-point weekly return spread between extreme short and extreme long positioning quartiles. Conversely, copper and natural gas exhibit momentum-like behaviour at the extremes, driven by physical supply-demand imbalances.

5.2 Practical Implications for Portfolio Management

Beyond the econometric outputs, these findings offer highly practical, “down-to-earth” guidelines for real-world investors. The core takeaway is that blindly investing in a broad, equally-weighted commodity index is a flawed strategy that can actively damage a portfolio during crises.

Including commodities in a portfolio offers profound benefits only if the assets are tactically selected based on the macroeconomic environment. For downside protection, gold is the only reliable candidate: when equity markets crash and panic ensues, gold essentially ignores the stock market, preserving the investor’s capital precisely when they need liquidity the most.

Conversely, relying on industrial metals like copper, or dual-purpose metals like silver, to diversify equity risk is a dangerous misconception. Because their demand is tied to the global business cycle, they suffer from the “diversification paradox”: they crash exactly when the stock market crashes, amplifying portfolio losses rather than mitigating them. Similarly, highly volatile assets like natural gas may look attractive on paper due to massive episodic price spikes, but in practice, the constant and costly rebalancing required to manage their wild swings eats away at the investor’s real returns.

Ultimately, this research suggests that investors should abandon the idea of a generic “commodity allocation”. Instead, they should utilise precious metals strictly for structural insurance, avoid procyclical metals during recessionary bear markets, and actively monitor institutional positioning (COT data) to avoid buying into agricultural or metal assets when speculators are already dangerously overextended.

5.3 Limitations and Future Work

Several methodological limitations must be explicitly acknowledged. First, the Mean-Variance and CVaR portfolio optimisations are evaluated strictly *in-sample*, inherently incorporating look-ahead bias to map theoretical diversification boundaries rather than actionable out-of-sample trading performance. Second, the underlying GARCH(1,1) specification relies on Gaussian innovations, failing to fully capture the fat tails and asymmetric leverage effects prevalent in commodity returns. Third, the DCC estimations for silver and natural gas converged to corner solutions, suggesting the need for more flexible non-linear dependency models, such as asymmetric DCC or copula-based approaches. Finally, the machine-learning forecasts rely on a single chronological split rather than a robust expanding-window cross-validation.

Future research will address these constraints across three dimensions. First, we will transition from static in-sample optimisation to a rigorous *out-of-sample* walk-forward backtest, dynamically allocating capital based on the endogenous HMM regimes while explicitly accounting for the 5 bps transaction-cost penalty. Second, we will adopt asymmetric, fat-tailed variance specifications (e.g., GJR-GARCH with Student-*t* innovations) to better model conditional volatility. Lastly, we aim to incorporate physical-market fundamentals, such as LME inventory levels, USDA crop reports, and EIA storage draws, into the Machine Learning feature space to test whether physical supply-side imbalances can enhance tactical timing beyond purely macro-financial signals.

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